

Budget Justification

Personnel Costs:

Senior/Key Personnel

PI: Steven Anton, Associate Professor of Mechanical Engineering, will dedicate 3.5 months of his 9-month academic appointment and one summer month annually to this project. Serving as the inaugural Director of the Institute of Biomedical Research (IBR), Dr. Anton will provide the critical leadership required for the Institute's establishment and initial growth, an effort necessitating a 50%-time commitment. In addition to his leadership of the IBR, PI Anton will coordinate BRE-SPAD initiatives focused on faculty training and development and manage the Pilot Research Project Program (PRPP). Furthermore, he assumes full responsibility for all PI/PD obligations, including mandated training, stakeholder meetings, and the rigorous evaluation of program goals and outcomes.

PI: John Liu, Interim Provost and Professor of Biology, will provide institutional oversight for the BRE-SPAD project. As a 12-month administrator, he will contribute his expertise at no cost to the grant. His participation aligns with his core mission to enhance TTU's biomedical research infrastructure. Dr. Liu will dedicate at least one month of effort annually to lead institutional policy changes, cluster hiring, and faculty workload optimization. Furthermore, alongside Mr. Hunt, he will facilitate and manage high-level strategic partnerships with Vanderbilt University and the University of Tennessee, Knoxville (UT) to broaden the project's impact and resource network."

PI: Neal Hunt, Assistant VP for Research Administration, will lead the enhancement of the university's sponsored program infrastructure. Leveraging his extensive expertise in research administration, Mr. Hunt will work alongside Dr. Liu to manage strategic partnerships with Vanderbilt and UT. PI Hunt will commit one month of effort annually during Years 1 and 2 to launch key initiatives, including staff recruitment, professional training, and a strategic staff exchange program with our partner organizations. These foundational activities are designed to be fully established and implemented within the first two years of the grant. While no formal effort is charged for Years 3–5, Mr. Hunt will provide ongoing leadership to ensure the continued growth of sponsored programs and the sustainability of our partnerships with Vanderbilt and UT.

PI: Xuanzhi Zhan, Associate Professor of Chemistry, will serve as Director of the Biomedical Research Enhancement Program (BREP), managing student recruitment, engagement, and longitudinal training. Dr. Zhan has extensive experience in coordinating undergraduate research, curriculum-based research integration, and advising student research, especially undergraduate students. PI Zhan will commit 1 academic month and 2

summer months to this work in Years 1 and 2. During this phase, he will be dedicated to strategically establish the campus-wide undergraduate BREP program and develop the proposed activities regarding student trainee recruitment, training, mentored-research program, and career development. During Years 3-5, the framework of the BREP will be fully developed. Therefore, PI Zhan will commit 1 academic month and 1 summer month to this work in Years 3-5, and will continue to oversee all efforts related to the Biomedical Research Enhancement Program (BREP).

Other Personnel

Staff Specialist for NIH programs within ORED, the Office of Research and Economic Development (ORED) has committed to hiring one new pre-award specialist to become an expert on NIH and other biomedical funding opportunities. This position will be co-sponsored by ORED and the BRE-SPAD-TTU program. A total of 0.5 FTE (6 months of effort, \$25,000 annually) is budget for this position annually throughout the five-year project period.

Program Evaluator: Dr. Krista Davis, Director of Research and Evaluation at Metro Nashville Public Schools, will serve as the External Program Evaluator for the BRE-SPAD project. As an independent evaluator with specialized expertise in grant-funded program assessment, Dr. Davis will lead the evaluation design, oversee data collection protocols, conduct quantitative and qualitative analyses, and prepare both annual and final performance reports. A total of \$12,000 is budgeted for Year 1, reflecting the increased effort required to establish the comprehensive evaluation framework and strategic assessment tools. For Years 2–5, an annual budget of \$10,000 is allocated to support her ongoing data monitoring, analysis, and reporting responsibilities.

Student Assistants. Funding is requested to support two student assistants.

- **Graduate Assistant for Program Evaluation:** Funding is requested for one Graduate Assistant (GA) to support the External Evaluator, Dr. Krista Davis. The GA will serve as a vital liaison, coordinating data collection protocols, managing database access, and providing administrative support for the evaluation framework. Support is requested at a rate of \$20/hour for 120 hours annually, totaling \$2,400 per year.
- **Undergraduate Intern for the Institute of Biomedical Research (IBR):** Funding is requested for one Undergraduate Intern to support the IBR Director, PI Anton. This intern will assist with the daily operational management of the Institute, including coordinating BRE-SPAD scheduled activities, maintaining program documentation, and assisting with the communication of IBR initiatives to the campus community.

Support is request at a rate of \$15/hour for 10hours/week and 40 weeks per year, totaling \$6,000 per year.

Mentorship for PI Anton: Dr. Eric Ledet, Director of the Severino Center for Technological Entrepreneurship and Professor of Biomedical Engineering at Rensselaer Polytechnic Institute, will commit 0.5 summer months to this work each year. Dr. Ledet, as an expert in biomedical research in orthopedics with funding from NIH and experience as an institute director, will advise Dr. Anton in institute leadership and research administration. Given their similar research backgrounds in orthopedics and biomedical sensing, Dr. Ledet is an ideal mentor for PI Anton.

Mentorship for PI Liu and PI Hunt: Dr. Harris Lewin, Research Professor at Arizona State University, will commit equivalent of 0.5 months to this work each year. He has extensive experience in growing research enterprise in various capacities, including as Director of the Institute of Genome Biology at University of Illinois, and Vice Chancellor for Research at UC-Davis. As a member of the National Academy of Sciences, Dr. Lewin will provide advice based on the expertise pool of our existing faculty and new faculty lines related to biomedical research.

Mentorship for PI Zhan: Dr. Carolyn Roberts, Senior Director of Immersion Vanderbilt, will serve as a formal mentor to Dr. Zhan, focusing on the strategic management of the BREP and the cultivation of a student-centered undergraduate research ecosystem. Dr. Roberts will dedicate 0.5 months of effort annually to this partnership. Leveraging her proven track record in establishing and leading campus-wide research enhancement initiatives at Vanderbilt University, she will provide critical guidance on program scalability, student engagement, and the successful implementation of the BREP framework at Tennessee Tech.

Fringe benefit rates: Fringe benefit rates for TTU employees are calculated based on the policy of TTU at 42% for academic months and 29.6% for summer months.

Major Equipment Request:

Based on a comprehensive shared-instrument inquiry led by the PI team, two critical instruments have been identified to support the research activities of faculty across the Departments of Biology, Chemistry, and Chemical Engineering.

- **BZX Fluorescence Microscope (BZ-X1100) [\$67,441]:** This high-resolution imaging system is essential for the proposed biomedical research activities, particularly for cellular analysis and material characterization.

- **Odyssey XF Imaging System (9150-01) [\$67,890]:** This system will provide versatile, high-sensitivity imaging for western/southern blots and other biochemical assays.

These instruments will be housed in a shared facility to maximize access for BRE-SPAD trainees and faculty, fostering a collaborative, cross-departmental research environment. Formal vendor quotes for the BZX Fluorescence Microscope and the Odyssey XF Imaging System are included as appendices to this justification for review.

Travel:

Funds are requested for various program-related travel, as follows:

- PIs to attend the required annual BRE-SPAD meeting. We understand that this may not be enough for all four PI's, but for two a year, and the PD's can go to such meetings in alternative years.
- Support biomedical research faculty to attend biomedical research conferences, travel to meet NIH program directors, and/or travel to meet external research collaborators. We anticipate these funds to be sufficient to support four trips per year.
- Support students (undergraduate or graduate) to attend biomedical conferences. Students obtaining this funding will be required to present their research results at the conference (i.e., poster, oral, etc.). We anticipate these funds to be sufficient to support four trips per year because registration is generally less expensive for students than for faculty.
- Support travel for invited speakers to come to campus, deliver seminars, and collaborate and mentor our faculty and students. We anticipate supporting up to three trips per year.
- Additional travel funds are requested to support two ORED staff members to attend the Society of Research Administrators International (SRAI) Annual Meeting and register for the pre-conference "Introduction to the National Institutes of Health" training workshop. ORED staff members will complete SRAI's "National Institutes of Health Grants Fundamentals" Certificate program during the conference by attending the required sessions. If additional training is needed, ORED will support additional training within their regular budget.

Other Direct Costs:

Consultant Fees

A total of \$205,000 is requested over the five-year project period for mentors at our partner organizations. These fees are tiered to reflect the intensive program design and infrastructure launch required in the initial phase: \$50,000 in Year 1 for foundational strategic development, followed by \$40,000 annually in Years 2-4 for implementation oversight. As the program reaches operational maturity, the budget scales to \$35,000 in Year 5 to support final evaluations and sustainability planning. These consultants will provide critical expertise in biomedical research enhancement and institutional capacity building. We anticipate these fees will also be used for travel from the partner organizations to Tennessee Tech and vice versa (~100 miles from Cookeville to Nashville and from Cookeville to Knoxville).

Biomedical Research Enhancement Program (BREP)

A total of \$87,000 (\$28k Year 1, \$20k Year 2, \$15k Year 3, \$12k Year 4, \$12k Year 5) are requested to fund the various initiatives proposed under the Biomedical Research Enhancement Program (BREP). We anticipate the development of BREP initiatives to be more heavily focused in beginning years, hence a tapered funding plan. Examples of key funded BREP initiatives include:

Curriculum Development: We are budgeting \$5k/course for supplemental pay. We anticipate at least 2 new courses being developed. This is budgeted in Year 1 so that new courses are developed early in the BRE-SPAD program and can be taught for the duration of the project period.

BREP Biomedical Research Grant program: Awards (up to \$3,000) are granted to student research teams to support their BREP mentored project or capstone projects.

Workshops: Funds are requested to support faculty time (primarily in Year 1) to develop various student-focused workshops. Topics include research skills training, writing a CV, graduate school information session, and careers in industry.

Research Lab Mingle event: Funds are requested to support an annual social mixer event to enhance the research environment. The budget covers food.

Biomedical Innovation Competition: Funds are requested for venue costs, poster printing costs (so that students will incur no cost to attend the event), and up to \$4k of student awards.

Pilot Research Project Program (PRPP)

The single largest funding is requested for the proposed PRPP. We are requesting a total of \$680,000 over the project duration to fund the PRPP (\$140k Year 2; \$180k for Years 3-5). We anticipate to fund 3-6 projects per year.

Mid-TN Biomedical Symposium

Funding is requested to establish and host the annual Mid-TN Biomedical Symposium starting in Year 2 (\$5,000 per year for four years, a total of \$20,000). This regional event will serve as a primary platform for BRE-SPAD-TTU trainees to present their research findings to peers and faculty from partner institutions. The budget covers venue costs, food, and keynote speaker honoraria, fostering a robust biomedical research community in Middle Tennessee.

Biomedical Research Seminar and Social Hour

A total of \$2,000 per year is budgeted to support a seminar series (3-4 times per semester) and professional networking hour hosted by the Institute of Biomedical Research (IBR). These sessions will bring together students and faculty from diverse disciplines to discuss emerging trends in biomedical science, fostering the cross-departmental collaboration essential to the program's success. The budget covers venue costs and food.

Training Costs

Funds of ~\$1400/year (with an annual increase of \$50/yr) are requested to support two ORED staff each year to register for SRAI training courses specific to NIH. Staff members may repeat training every few years, as necessary. We anticipate nearly all ORED grant specialists to have the opportunity to get trained with this funding.

Indirect Costs

Indirect costs are requested at a rate of 43.5% of the direct cost, the negotiated indirect cost rate as approved by DHHS.



Quotation

Attn : Xuanzhi Zhan

Dept :

Fax :

Email : xzhan@tntech.edu

Quoted to : Tennessee Tech Univ.

115 West 10th Street Mechanical Engineering Dept.

COOKEVILLE, Tennessee 38505

Quotation : 13789231

Quote Date : 1/22/26

Your Reference :

Thank you for the opportunity to provide you a quotation for the following product(s).

***Please reference the Keyence Quotation number 13789231 when placing your order!

Reference :

Sales Contact : GIBSON, ASHTON (TMP)

Tel : 1-888-KEYENCE Ext :

Fax : 855-539-0123

Email: Ashton.Gibson@keyence.com

Line	Item	Item Description	Taxable	Unit Price	Qty	Ext Price
			Unit			
1	BZ-X1100	BZX Fluorescence Microscope	pcs		1	
		BZX Microscope Head				
	HS Code: 9011800000					
2	BZ-X1000	BZX Fluorescence Microscope	pcs		1	
		BZX Microscope Controller				
	HS Code: 8471801000					
3	BZ SUPPORT PACKAGE	BZ Support Package	pcs		1	
	HS Code:					
4	BZ-A1W	BZX Accessory	pcs		1	
		BZX Standard Accessory Package				
	HS Code: 8523492010					
5	OP-89167	BZX Accessory	pcs		1	
		BZX Multi-band set				
	HS Code: 9001200000					
6	BZ-PF04P	BZ Objective Lens (4x Phase)	pcs		1	
		Plan Fluorite 4x PH				
	HS Code: 9002909500					
7	BZ-PF10P	BZ Objective Lens (10x Phase)	pcs		1	
		Plan Fluorite 10x PH				
	HS Code: 9002909500					
8	BZ-PF20LP	BZ Objective Lens (20x Phase)	pcs		1	
		Plan Fluorite 20x LD PH				
	HS Code: 9002909500					
9	BZ-PF40LP	BZ Objective Lens (40x Phase)	pcs		1	
		Plan Fluorite 40x LD PH				
	HS Code: 9002909500					



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Reference :

Sales Contact : GIBSON, ASHTON (TMP)

Tel : 1-888-KEYENCE Ext :

Fax : 855-539-0123

Email : Ashton.Gibson@keyence.com

Line	Item	Item Description	Taxable	Unit Price	Qty	Ext Price
			Unit			
10	OP-89156	BZX Accessory			1	
		BZX Fixed holder frame	pcs			
	HS Code:	8471801000				
11	OP-89157	BZX Accessory			1	
		BZX Sample holder set_Slides/Dishes	pcs			
	HS Code:	8471801000				
12	OP-89158	BZX Accessory			1	
		BZX General purpose frame set	pcs			
	HS Code:	8471801000				



Quotation

Attn : Xuanzhi Zhan

Dept :

Fax :

Email : xzhan@tntech.edu

Quoted to : Tennessee Tech Univ.

115 West 10th Street Mechanical Engineering Dept.

COOKEVILLE, Tennessee 38505

Quotation : 13789231

Quote Date : 1/22/26

Your Reference :

Thank you for the opportunity to provide you a quotation for the following product(s).

***Please reference the Keyence Quotation number 13789231 when placing your order!

Reference :

Sales Contact : GIBSON, ASHTON (TMP)

Tel : 1-888-KEYENCE Ext :

Fax : 855-539-0123

Email: Ashton.Gibson@keyence.com

HS Codes provided for general information purposes only. Keyence is not responsible for any custom clearance issues if HS codes used by a third party for import purposes.

Sub Total	66,910.00
Misc. Charges	0.00
Shipping & Handling	531.00
Tax	0.00
Total (USD)	67,441.00

AVAILABILITY : Limited Qty In Stock

Shipping : SEKO LTL Ground

F.O.B Point : Itasca, IL

Terms : Terms Pending

Quote Expires: 02/21/26

Quoted shipping & handling cost is an estimate based on quantities and method of delivery.

Prices, Availability and Terms are subject to Change.

FOR ORDERS & CUSTOMER SUPPORT, PLEASE CONTACT:
CUSTOMER SERVICE DEPT.
CustomerService@Keyence.com or TEL: 888-539-3623

CBE Waste Recycling Fee is paid by Keyence
on behalf of California customers

FEDERAL ID NO: 87-3735190
DUNS NO: 119122920

LI-COR Biotech, LLC
 4647 Superior St.
 Lincoln, NE 68504 USA
 Phone: 402-467-0800
 U.S. & Canada: 800-645-4267
 www.licorbio.com
 bio.orders@licorbio.com

Dr. Paul (Anthony) Barnes
 Tennessee Tech University
 1100 N Dixie Road
 Cookeville, TN 38505
 UNITED STATES

QUOTE NO: L23632-1
QUOTE DATE: January 29, 2025

Quote Expires	Payment Terms	LI-COR Reference Number
February 8, 2025	0.00/0/30	
Ship Via	Shipping and Handling Terms	LI-COR Consultant
	Prepaid & Add	Tallie Patsilevas

Item No.	Qty	Shipment ARO	Part No. and Description	Unit Price	Net Price	Price Extension
1	1	30	9150-01 Odyssey F (2-Laser) Imaging Bundle - Basic	\$67,540.00	\$67,540.00	\$67,540.00

Comments:
 *INFORMATION FOR INSTRUMENT PURCHASES OVER \$15,000.
 Customers requiring inside delivery and liftgate service will need to specify such on the issued Purchase Order. Inside delivery and lift gate service will be invoiced at \$300.00. These costs will be in addition to any Shipping and Handling charges listed on this quote.

Subtotal:	\$67,540.00
Total Price:	\$67,540.00
Shipping and Handling (est.):	\$350.00
Grand Total:	\$67,890.00

Item No.	Product Details
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1	9150-01 Odyssey F (2-Laser) Imaging Bundle - Basic
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Odyssey F (2-Laser) Imaging Bundle - Basic
 Includes the following items:
 - Instrument: F (2-Laser) Infrared Imager
 - Acquisition Software: Image Studio
 - Data Analysis Software: Image Studio

- Accessories: F (2-Laser) Accessories

- Manuals: Owner manual

- Computer: Desktop PC and Monitor

- Warranty: One year

- Installation: Remote

Included with the package to generate the most robust and replicable data to meet current publication requirements (value greater than \$25,000):

- Remote Training: New user hardware, software, and application training with an Expert Coach. Contact LI-COR for availability of Remote training.

- Remote Advanced Training: Personalized advanced software and application training with an Expert Coach. Contact LI-COR for availability of Remote training.

- Applications Guides: Contains protocols optimized for meeting publisher requirements for Western blot and In-Cell Western data submission

- Acquisition Software: Image Acquisition software which collects raw, unaltered images.

- Data Analysis Software: Software which is based on publisher, granting agencies and industry experts best practices to ensure data robustness.

- New user startup reagents that have been optimized for the highest quality data

- Unlimited access to internal application and instrument experts via phone, e-mail or chat

- Unlimited access to online educational materials

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From LICORbio, Lincoln, Nebraska, USA Standard

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3. **Prices/Taxes.** All prices are quoted for delivery to Buyer when goods are loaded on the carrier at LICORbio premises in Lincoln, Nebraska, USA exclusive of shipping, insurance and installation charges, all of which are Buyer's sole responsibility. All prices are exclusive of all sales, use, excise, value added, withholding and other taxes, all customs, duties, documentation charges, and freights forwarder charges and charge of any nature now or hereafter claimed or imposed by any governmental authority upon the sale of the Products or performance of the Services. Any such charges will be added to the product invoice or subsequently invoiced to the Buyer. In the event LICORbio is required to pay any such tax, duty or charge, Buyer will promptly reimburse LICORbio.

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5. **Return Policy.** Buyer may return non-consumable Products to LICORbio within forty-five (45) days (Thirty (30) days if C-DiGit® Blot Scanner) of invoice date only with prior authorization by LICORbio. Temperature controlled Reagents are strictly non-returnable. The Product(s) being returned new and unused condition and must be resalable as new. Any returned Product(s) are subject to payment of a fifteen percent (15%) re-stocking fee on all items returned. This fee is not applicable for the

C-DiGit® Scanner. Buyer shall be responsible to make payment to LICORbio for any and all expenses related to de-installation of the Product(s), including but not limited to shipping, duties, and taxes. All payments subject to this provision shall be made to LICORbio within thirty (30) days of return, or de-installation, of the Product(s).

6. **Delays In Performance.** LICORbio shall not be liable for any delay in performance hereunder due to unforeseen circumstances or due to circumstances beyond its control including, but not limited to, acts of nature, acts of government, labor disputes, delays in transportation, delays in customs clearance and delays in delivery or inability to deliver by LICORbio suppliers.

7. **Shipment and Packing.** All Product prices exclude costs of shipping and handling and insurance, in accordance with delivery terms designated by LICORbio. Unless otherwise agreed in writing, such costs will be paid by the Buyer and will appear as a separate item on LICORbio invoice. LICORbio shall ship in accordance with LICORbio standard practices. Buyer may specify different shipping instructions, subject to agreement by LICORbio. Unless otherwise agreed to in writing by LICORbio, all products shall be packaged, if appropriate, for shipment and storage in accordance with standard commercial practices. All packing shall conform to carrier requirements.

8. **Partial Shipments.** LICORbio reserves right to make delivery in partial shipments ("Installments"). Any Products delivered in Installments may be invoiced individually and is payable subject to Section 4 of these Conditions. Additional shipping and handling charges for Installments may apply. Delay in delivery of any Installment shall not relieve Buyer of Buyer's obligation to accept remaining deliveries.

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11. **Acceptance.** All sales are final and all Products shall automatically be deemed accepted upon delivery to Buyer when goods are loaded on the carrier at LICORbio premises in Lincoln, Nebraska, USA. Failure to provide written notice to LICORbio of any shortages, defects, or damages relating to the Products within fifteen (15) days after receipt shall conclusively deem that the Products conform to the terms set forth in these Conditions. Buyer may not return any Products to LICORbio except as provided for by LICORbio warranty or as provided herein.

12. **Product Warranties.** Unless otherwise specified by LICORbio:

(a) LICORbio warrants that, for a period of twelve (12) months from the date of shipment of the Products from LICORbio (the "Warranty Period"), unless otherwise specified for individual Products (such as Reagents with

a specified shelf life) or extended by a Support Contract or Extended Warranty Contract (the "Extended Warranty"), the Products sold hereunder will be free from material defects in materials and workmanship and will conform to LICORbio published specifications in effect as of the date of manufacture. LICORbio SPECIFICALLY DISCLAIMS ANY INDIRECT, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES (INCLUDING LOSS OF USE OR LOST PROFITS) WHICH MAY RESULT FROM THE USE OF PRODUCTS PURCHASED HEREUNDER, AS FURTHER SET FORTH IN SECTION 13 OF THESE CONDITIONS OF SALE. This limited warranty extends only to Buyer as original purchaser unless otherwise agreed upon in writing by LICORbio.

(b) The foregoing warranty/extended warranty coverage shall not apply if the defective Product (i) has been subjected to abuse, misuse, neglect, negligence, accident, improper testing, improper installation, improper storage, improper handling or use contrary to any instructions issued by LICORbio, (ii) has been repaired or altered by persons other than LICORbio, (iii) has been moved/relocated once originally installed unless LICORbio approved deinstall/reinstall procedures are followed; (iv) has not been installed, operated, repaired and maintained in accordance with the documentation or operated outside of the environmental specifications for the Product; (v) has failed due an Act of God, including but not limited to fire, flood, tornado, earthquake, hurricane or lightning or (vi) has been used with any devices, accessories or products not manufactured by or approved by LICORbio. In addition, the foregoing warranty shall not apply to Products (i) marked or identified as "sample," (ii) loaned or provided to Buyer at no cost, or (iii) which are sold "as is."

(c) If during the Warranty/Extended Warranty Period: (i) LICORbio is notified promptly in writing upon discovery of any defect in the Product, including a detailed description of such alleged defect, (ii) such Product is returned, transportation charges prepaid, to LICORbio designated manufacturing facility subject to the prior approval of LICORbio with a valid Return Material Authorization ("RMA") number, and (iii) LICORbio inspections and tests determine that the Product is indeed defective and the Product has not been subjected to any of the conditions set forth above, then, as Buyer's sole remedy and LICORbio sole obligation under the foregoing warranty, LICORbio will, at LICORbio option, repair or replace without charge the defective Product. In no event will the Buyer itself nor will the Buyer allow any party other than LICORbio or a third party authorized in writing by LICORbio to perform any service on the Products.

(d) During the Warranty/Extended Warranty Period, LICORbio will provide on-site warranty repair for certain Odyssey® family Imagers, or the Pearl® Imager, including travel costs, repair parts, and labor to maintain the hardware in proper operating condition. Some Products such as the C-DiGit® and D-DiGit® scanners and the Odyssey® M imager, may have failures can only be repaired at the factory. In such cases on-site warranty repair will not be provided by LICORbio. At LICORbio discretion, the Buyer may be required to run certain diagnostic procedures to help determine the source of the problem before on-site warranty repair is rendered. If an on-site service call is initiated, LICORbio will dispatch a service technician to the Buyer site. On-site service will be provided 8:00 a.m. to 5:00 p.m. (Buyer local time), Monday through Friday, excluding LICORbio holidays. The cost of a repair/service call for an instrument malfunction caused by third party hardware and/or software will be billed to Buyer on a time and material basis.

(e) Any Product that has either been repaired or replaced under this warranty shall have warranty coverage (parts only) for the longer of one (1) year or the remaining original warranty period. Replacement parts and/or replacement Products used in the repair or replacement of Products may be new or equivalent to new at LICORbio sole discretion.

(f) EXCEPT FOR THE WARRANTIES SET FORTH IN THIS SECTION, LICORbio MAKES NO OTHER WARRANTIES, EXPRESS, IMPLIED OR STATUTORY, WITH RESPECT TO ANY SERVICES, PRODUCTS OR OTHER PRODUCTS PROVIDED IN CONNECTION WITH THESE

CONDITIONS, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTY OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, NONINFRINGEMENT, OR ARISING FROM COURSE OF PERFORMANCE, DEALING, USAGE OR TRADE.

(g) Notwithstanding anything herein to the contrary, LICORbio makes no warranty with respect to any third party products provided under these Conditions. Buyer's sole remedy with respect to such third party products shall be pursuant to the original manufacturer's or licensor's warranty, if any, to Buyer, to the extent permitted by the original manufacturer or licensor.

13. Limitation of Liability. IN NO EVENT SHALL LICORbio, ITS LICENSORS OR ITS SUPPLIERS BE LIABLE TO BUYER OR ANY THIRD PARTY FOR COSTS OF PROCUREMENT OF SUBSTITUTE PRODUCTS OR SERVICES, LOST PROFITS, DATA OR BUSINESS, OR FOR ANY INDIRECT, SPECIAL, INCIDENTAL, EXEMPLARY OR CONSEQUENTIAL DAMAGES OF ANY KIND ARISING OUT OF OR IN CONNECTION WITH THE USE OF THE PRODUCTS OR THESE CONDITIONS, HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY (WHETHER IN CONTRACT, TORT (INCLUDING NEGLIGENCE), STRICT LIABILITY, PRODUCTS LIABILITY OR OTHERWISE). LICORbio TOTAL AND CUMULATIVE LIABILITY ARISING OUT OF OR IN CONNECTION WITH ANY PRODUCTS PURCHASED BY BUYER OR SERVICES PERFORMED BY LICORbio ON BEHALF OF BUYER HEREUNDER SHALL IN NO EVENT EXCEED THE PURCHASE PRICE PAID BY BUYER FOR SUCH PRODUCTS OR SERVICES. THE LIMITATIONS SET FORTH IN THIS SECTION SHALL APPLY EVEN IF LICORbio OR ITS SUPPLIERS HAVE BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES, AND NOTWITHSTANDING ANY FAILURE OF ESSENTIAL PURPOSE OF ANY LIMITED REMEDY.

14. Authorized Use of Products. Unless otherwise expressly indicated in LICORbio catalogues, LICORbio website or on the label or other documentation accompanying Products, the LICORbio Products are intended for RESEARCH USE ONLY and are not to be used for any other purposes including, but not limited to, unauthorized commercial purposes, in vitro diagnostic purposes, ex vivo or in vivo therapeutic purposes, investigational use, in foods, drugs, devices or cosmetics of any kind, or for consumption by or use in connection with or administration or application to humans or animals. Buyer acknowledges that the Products have not necessarily been tested for safety or efficacy, unless expressly stated in LICORbio catalogues or on the label or other documentation accompanying the Products.

15. Severability. If any portion of these Conditions is held invalid, the parties agree that such invalidity shall not affect the validity of the remaining portions of these Conditions.

16. Export Control. Buyer acknowledges and agrees that the Products purchased under these Conditions or Services performed by LICORbio may be subject to restrictions and controls imposed by the United States Government and the regulations thereunder. BUYER WARRANTS THAT IT WILL NOT EXPORT OR RE-EXPORT ANY PRODUCTS PURCHASED OR DELIVERABLES FROM SERVICES PERFORMED BY LICORbio WITHOUT PRIOR WRITTEN NOTIFICATION AND APPROVAL OF LICORbio.

17. Assignment. Buyer shall not assign or transfer these Conditions or any rights or obligations under these Conditions, whether voluntary or by operation of law, without the prior written consent of LICORbio. LICORbio may assign or transfer these Conditions to any successor by way of merger, acquisition or sale of all or substantially all of the assets relating to these Conditions. LICORbio or any successor may assign all or part of the right to payments under these Conditions. Any assignment or transfer of these Conditions made in contravention of the terms hereof shall be null and void. Subject to the foregoing, these Conditions shall be binding on and inure to the benefit of the parties' respective successors and permitted assigns.

18. **Entire Agreement.** These Conditions of Sale and Performance of Services take precedence over Buyer's additional or different terms and conditions, to which notice of objection is hereby given. Acceptance by Buyer is limited to LICORbio Conditions of Sale. Neither LICORbio commencement of performance nor delivery shall be deemed or construed as acceptance of Buyer's additional or different terms and conditions. These Conditions supersede all prior communications, transactions, and understandings, whether oral or written, and constitute the sole and entire agreement between the parties pertaining to the referenced quotation or purchase order, provided that: (1) these Conditions shall not, without LICORbio prior written consent, supersede any conflicting terms of: (a) prior written agreements duly executed by LICORbio, or (b) governmental purchase orders, terms of purchase, requests for quotation or acquisition regulations relative to governmental purchasers; and (2) to the extent not in conflict with any such prior or governmental terms, these Conditions shall supplement them. No modification, addition or deletion, or waiver of any of the terms and conditions of these Conditions shall be binding on either party unless made in a non-preprinted agreement clearly understood by both parties to be a modification or waiver, and signed by a duly authorized representative of each party.

19. **Force Majeure.** Shipping/delivery dates are approximate and may be delayed absent prompt receipt from Buyer of all necessary information. LICORbio shall not be responsible for any failure to perform or delay attributable in whole or in part to any cause beyond its reasonable control, including but not limited to Acts of God, government actions, war, civil disturbance, insurrection, sabotage, labor shortages or disputes, failure or delay in delivery by LICORbio suppliers or subcontractors, transportation difficulties, customs clearance, shortage of energy, raw materials or equipment, or Buyer's fault or negligence. In the event of any such delay the date of delivery shall, at the request of LICORbio, be deferred for a period equal to the time lost by reason of the delay.

20. **Governing Law and Venue.** These Conditions and performance by the parties hereunder shall be construed in accordance with the laws of the State of Delaware, U.S.A., without regard to provisions on the conflicts of law.